Current makes a magnetic field

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect current direction with the magnetic field around a wire.

SUCCESS CRITERIA - I can...

1. Sketch circular field lines around a straight wire.
2. Use a stated right-hand rule.
3. Predict the effect of reversing current.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “current” and “magnetic field” to explain your answer. Sentence starters are optional.

current: rate of charge flow

magnetic field: a field that influences magnets and moving charges

radius: distance from the center to the edge of a circle or sphere

01 Warm-up: bring back an earlier idea

A **6 Ω** resistor is in series with two parallel **8 Ω** resistors across **20 V**. **Find parallel equivalent**, total resistance and supply current.

Word help: series = connected along one unbranched current path

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $B = \mu_0 I/(2\pi r)$; $\mu_0 = 4\pi \times 10^{-7}$ **T·m/A**.
Long straight wire. Right thumb follows conventional current; curled fingers show field direction.

Worked reasoning: Around a straight wire carrying conventional current out of the page, the magnetic field circles counterclockwise by the right-hand rule. A dot means out of page.

Your turn: Draw three field arrows around the wire and **explain what the dot represents**.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Current out of page (dot). Add field arrows.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Conventional current points out of the page. Use the right-hand rule to **draw the surrounding magnetic field** direction.

Sentence starter: In my work, I used … to show … . I labeled it … .

Current out of page (dot). Add field arrows.

Marzano check - circle: ① ② ③

04 Apply the physics

A long straight wire carries **5.0 A** out of the page. **Calculate B** at **0.020 m** using $B = \mu_0 I/(2\pi r)$, $\mu_0 = 4\pi \times 10^{-7}$ **T·m/A**. **Mark the location** and local field direction on the diagram.

Sentence starter: I found … [unit]. This value means … in this situation.

Current out of page (dot). Add field arrows.

Marzano check - circle: ① ② ③

Current makes a magnetic field

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

What happens to field direction if the current reverses while its magnitude stays the same?

Word help: magnitude = size of a quantity without its direction

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A neutral object is attracted to an object carrying **+4 nC**.
Explain how attraction can occur without giving the neutral object a net charge.
Label charges and both interaction forces.

Word help: charge = a property responsible for electric interactions

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .



Charge A Charge B

Marzano check - circle: 1 2 3

07 Exit check

Current triples and distance doubles. Express new B as a multiple of the old value.

Sentence starter: I notice … . This supports … because … .

Current out of page (dot). Add field arrows.



Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

For a long straight wire use $B=\mu_0 I/(2\pi r)$, with $\mu_0=4\pi\times10^{-7}$ **T·m/A**. **Find B** for **5 A** at **0.10 m**; then double distance.

Sentence starter: I found … [unit]. This value means … in this situation.

Cross = current into page. Add magnetic-field arrows.



Marzano check - circle: 1 2 3